Ball lifts of the split Cayley hexagon: bipartite graphs of diameter 9, 10 and 11

Degree–diameter lab (Claude Code sessions coordinated by Doğa Özden)

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Draft of 28 October 2026 (job thm-polygon-levels; branch `claude/haiku-squads-diameter-7-hr64vz` of github.com/dogaozden/degree-diameter). Written in the style of the lab's diameter-7 note (research/theorems/D7_HEXAGON.md on branch `claude/d7-hexagon-theorems-4w1prt`, "the D7 note"), whose coordinates and notation are used. Every claim is labelled **Theorem** (proved here, by hand), **Computer-checked** (a statement about specific finite objects, with the check named) or **Heuristic** (an argument with a gap that is named).

0. Results

The diameter-7 level graphs (copies of the incidence graph of $H(q)$ joined by matchings, the 2013 table's $D = 7$ column) let a walk change layers only at its first or last vertex. Here the jump may start anywhere in a ball of radius ρ around the walk's end, and the layer it reaches depends on the **colour class** of the jumping element. This gives one theorem for all diameters $c + 2 + \rho$:

- **Theorem 2.3 (ball lift).** For any bipartite graph I with point–line distances $\leq c$, k balanced ρ-dominating colour classes on each side give a $\mathbb{Z}_{2rk+1}$-lift of I plus r matchings, of diameter $\leq \rho + c + 2$. The classical level graph is the case $\rho = 0$, $k = 1$.
- $D = 9$ **(Corollary 3.1):** $2N(q)(2rk+1)$ vertices, degree $q+1+r$, where k = number of balanced classes meeting every plane π_p (points) and every line 2-ball (lines). $k \leq q^2 + q + 1$, with equality iff $H(q)$ has a partition into ovoids and one into spreads (Proposition 3.2); $H(3)$ has no three pairwise disjoint spreads (Computer-checked 3.3). At $k = q^2 + q + 1$ the construction would beat the lab's value at every $(\Delta, 9)$, $7 \leq \Delta \leq 20$, except $\Delta = 8$; the record thresholds need k between 49% and 97% of q^2+q+1. Local search found balanced partitions with $k = 10$ at $q = 3$ and $k = 13$ at $q = 4$, and one-sided ones with $k = 15$ at $q = 5$ and $k = 20$ at $q = 7$ (77%, 62%, 48%, 35% of $q^2 + q + 1$; Computer-checked 3.4). That is far below the thresholds, so **no record at** $D = 9$.
- $D = 10$ **(Corollary 4.4):** for every q, a Hermitian spread gives $2N(q)(2r'(q^3+1)+1)$ vertices at degree $q + 1 + 2r'$, with no search. It reaches 84–97% of the lab's values at $(16, 10)$, $(19, 10)$, $(20, 10)$ and beats none.
- $D = 11$ **(Corollaries 5.2, 5.5):** K pairwise disjoint totally singular lines of $Q(6, q)$ (any line of the quadric is 4-dominating, Lemma 5.4) and K pairwise non-collinear points of $H(q)$ give $2N(q)(2rK + 1)$ vertices at degree $q + 1 + r$; for $q = 3^h$ the duality makes the second condition automatic. **New value at** $(10, 11)$**.** In $H(8)$ a cyclic group of order $63 = q^2 - 1$ of $G_2(8)$ has a union of 67 point orbits that is a set of 4,041 pairwise non-collinear points (97% of the upper bound 4,161), and $Q(6, 8)$ has 3,285 pairwise disjoint totally singular lines (55 orbits of the same group), 3,292 after a local search. With $K = 3{,}292$ and $r = 1$, Corollary 5.5 gives a bipartite 10-regular graph of diameter 11 with **493,203,330** vertices, against the lab's 445,709,568 (Theorem + Computer-checked 5.8). $(9, 11)$ stays open ($K = 1{,}743$ at $q = 7$, 1,852 needed). **First values** at $(\Delta, 11)$ for $11 \leq \Delta \leq 20$, where the lab has none, are in §5.7 (from $1.33 \cdot 10^9$ at $\Delta = 11$ to $3.9 \cdot 10^{11}$ at $\Delta = 20$).
- $D = 8$**, quadrangles, the Ree–Tits octagon (§6):** no cells.

All theorems were tested by building the graphs at small q and running an all-pairs BFS (§8).

1. Notation

Throughout, q is a prime power and $H = H(q)$ is the split Cayley hexagon in the coordinates of the D7 note, §1.1: $V = \mathbb{F}_q^7$ with coordinates $(X_0, \dots, X_6)$, the parabolic quadric

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2,$$

with polar form $b(X,Y) = Q(X+Y) - Q(X) - Q(Y)$. The points of H are the N points of $Q(6,q)$,

$$N = N(q) = \frac{q^6-1}{q-1} = (q+1)(q^4+q^2+1) = (q^3+1)(q^2+q+1),$$

the lines are the N lines of $Q(6,q)$ whose Grassmann coordinates satisfy Tits' equations $p_{12} = p_{34}$, $p_{54} = p_{32}$, $p_{20} = p_{35}$, $p_{65} = p_{30}$, $p_{01} = p_{36}$, $p_{46} = p_{31}$, and incidence is containment. P and L denote the point and line sets.

$I = I(q)$ is the incidence graph of H (vertex set $P \cup L$, $2N$ vertices, $(q+1)$-regular, bipartite), d its distance. H is a generalized hexagon of order (q,q), so (D7 note §1.2):

- (F1) I has diameter 6 and girth 12; point–line distances are $1, 3, 5$ and same-type distances $0, 2, 4, 6$.
- (F2) if $d(x,y) \le 5$ there is a unique shortest path from x to y.

For a point x, the lines of H through x are the $q+1$ lines through x of a totally singular plane π_x of $Q(6,q)$, and the points at distance ≤ 2 from x in I are exactly the q^2+q+1 points of π_x (each line through x contributes its q further points).

Balls. For an element x and an even $\rho \ge 0$, $B_\rho(x)$ is the set of elements **of the same type** as x at distance $\le \rho$ from x. So $B_0(x) = \{x\}$; $B_2(p) = \pi_p$ for a point p; $B_2(l)$ is l together with the $q(q+1)$ lines meeting l; and $|B_2(x)| = q^2+q+1$ for every x. A set A of elements of one type is **ρ-dominating** if it meets $B_\rho(x)$ for every element x of that type.

Ovoids and spreads. An ovoid of H is a set of q^3+1 pairwise opposite points (distance 6); a spread is a set of q^3+1 pairwise opposite lines. Every element is at distance ≤ 2 from exactly one element of an ovoid (resp. spread) of its type: two elements of an ovoid cannot share a 2-ball (they would be at distance ≤ 4), and $(q^3+1)(q^2+q+1) = N$. Spreads exist for every q (the Hermitian spread: the lines of H contained in a hyperplane section $Q^-(5,q)$); ovoids exist when q is a power of 3 (then $H \cong H^D$ and a duality maps a spread to an ovoid); see §9 for non-existence results.

2. The ball lift

The construction works for any bipartite graph in place of I; the hexagon enters only through the numbers in Corollaries 3.1, 4.4, 5.2 and 5.5. Fix:

- a connected bipartite graph I with colour classes P ("points") and L ("lines"), $|P| = |L| = N$, in which every point–line distance is at most an odd number c;
- an even $\rho \ge 0$ and integers $k, r \ge 1$;
- partitions $P = A_0 \cup \dots \cup A_{k-1}$ and $L = B_0 \cup \dots \cup B_{k-1}$ with $|A_i| = |B_i|$, such that every A_i is ρ-dominating (meets $B_\rho(p)$ for every point p) and every B_i is ρ-dominating (meets $B_\rho(l)$ for every line l);
- bijections $\sigma_1, \dots, \sigma_r : P \to L$ with $\sigma_t(A_i) = B_i$ for all t, i.

Put $m = 2rk+1$ and, for $z \in A_i$ and $1 \le t \le r$, $e_t(z) = (t-1)k + i + 1 \in \{1, \dots, rk\}$.

Definition 2.1 (the ball lift $\Lambda = \Lambda(I; A, B, \sigma; r)$). The vertex set is $(P \cup L) \times \mathbb{Z}_m$. The neighbours of a point vertex (z,g), $z \in A_i$, are

$$(l,g) \ \ (l \ni z), \qquad (\sigma_t(z),\ g + (t-1)k + i + 1) \ \ (1 \le t \le r);$$

the neighbours of a line vertex (l,g), $l \in B_i$, are

$$(p,g) \ \ (p \in l), \qquad (\sigma_t^{-1}(l),\ g - (t-1)k - i - 1) \ \ (1 \le t \le r)$$

(note $\sigma_t^{-1}(l) \in A_i$, so the two lists describe the same edges). In words: m copies ("layers") of I, and r perfect matchings between points and lines, the edge at z of the t-th matching climbing $e_t(z)$ layers.

Lemma 2.2 (Theorem). Let Γ be a connected bipartite graph with at least one edge. (a) If any two vertices of the same colour are at distance $\le 2s$, then $\operatorname{diam}\Gamma \le 2s+1$. (b) If any two vertices of different colours are at distance $\le 2s-1$, then $\operatorname{diam}\Gamma \le 2s$.

Proof. (a) Let u, v have different colours and let v' be a neighbour of v (it exists: Γ is connected with an edge). v' has the colour of u, so $d(u,v) \le d(u,v')+1 \le 2s+1$. (b) The same with u, v of the same colour. □

Theorem 2.3. Λ is a simple bipartite graph on $2Nm$ vertices in which every vertex has degree $\deg_I(x)+r$, and

$$\operatorname{diam}\Lambda \;\le\; \rho + c + 2.$$

Proof. Simple and bipartite. Every edge joins $P\times\mathbb{Z}_m$ to $L\times\mathbb{Z}_m$. An edge of I keeps the layer; the t-th matching edge at (z,g) changes it by $e_t(z) \in [(t-1)k+1,\, tk]$, which is never 0 modulo $m = 2rk+1$, and differs for different t. So no edge is repeated, and each vertex has $\deg_I(x)$ edges inside its layer and r matching edges.

Translations. $\tau_h : (x,g) \mapsto (x, g+h)$ is an automorphism, so it suffices to bound distances from vertices in layer 0. Write $F = \{1, \dots, rk\}$; then $\mathbb{Z}_m = \{0\} \cup F \cup (-F)$, a disjoint union, and every $h \in F$ is $(t-1)k+i+1$ for exactly one pair (t,i).

Two points. Let $p, p' \in P$ and $h \in \mathbb{Z}_m$; we show $d((p,0),(p',h)) \le \rho + c + 1$.

- $h = 0$: inside layer 0, $d_I(p,p') \le c+1$ (go from p to a line through p', then to p').
- $h = (t-1)k+i+1 \in F$: choose $z \in A_i \cap B_\rho(p)$. The walk $(p,0) \rightsquigarrow (z,0) \to (\sigma_t(z),h) \rightsquigarrow (p',h)$ has length $\le \rho+1+c$ (the middle edge is the t-th matching edge at z, which climbs $e_t(z) = h$ layers; the last segment is a point–line path of I inside layer h).
- $h = -((t-1)k+i+1) \in -F$: choose $z \in A_i \cap B_\rho(p')$. The walk $(p,0) \rightsquigarrow (\sigma_t(z),0) \to (z,h) \rightsquigarrow (p',h)$ has length $\le c+1+\rho$.

Two lines. Let $l, l' \in L$.

- $h = 0$: $\le c+1$ inside layer 0.
- $h = -((t-1)k+i+1)$: choose $w \in B_i \cap B_\rho(l)$; then $\sigma_t^{-1}(w) \in A_i$ and the walk $(l,0) \rightsquigarrow (w,0) \to (\sigma_t^{-1}(w),h) \rightsquigarrow (l',h)$ has length $\le \rho+1+c$.
- $h = (t-1)k+i+1$: choose $w \in B_i \cap B_\rho(l')$; the walk $(l,0) \rightsquigarrow (\sigma_t^{-1}(w),0) \to (w,h) \rightsquigarrow (l',h)$ has length $\le c+1+\rho$.

So any two vertices of the same colour are within $\rho+c+1$ (an even number). In particular Λ is connected (every vertex has a neighbour), and Lemma 2.2(a) gives $\operatorname{diam}\Lambda \le \rho+c+2$. □

Remark 2.4 (the classical case). $\rho = 0$, $k = 1$, $A_0 = P$, $B_0 = L$: the hypotheses hold trivially ($B_0(x) = \{x\}$) and Λ is the level graph with $2r+1$ layers and arbitrary matchings, of diameter $\le c+2$. For $H(q)$ ($c = 5$) this is Theorem A of the D7 note (the 2013 table's diameter-7 values $(2r+1)\cdot 2N(q)$); for a generalized quadrangle ($c = 3$) it is the diameter-5 polygon cover. The point of Theorem 2.3 is that a jump may start anywhere in a ball of radius ρ, so ρ more steps of diameter buy a factor $\approx |B_\rho|$ in the number of layers instead of a factor $\approx r$.

Remark 2.5 (relaxed data). The partitions can be replaced by pairwise disjoint ρ-dominating families $D_0, \dots, D_{k-1} \subseteq P$ and $E_0, \dots, E_{k-1} \subseteq L$ with $\sum_i \max(|D_i|, |E_i|) \le N$: enlarge the smaller of D_i, E_i by unused elements until $|D_i| = |E_i|$, then distribute the rest evenly (enlarging a dominating set keeps it dominating). This is the balance condition checked by the scorer `research/theorems/polylev/domcheck.py`.

Remark 2.6 (dualities). If φ is a duality of H (an incidence-preserving bijection $P \cup L \to L \cup P$ exchanging the types), then φ maps $B_\rho(p)$ onto $B_\rho(\varphi(p))$, so $\varphi(A_i)$ is a ρ-dominating set of lines with $|\varphi(A_i)| = |A_i|$. Hence when q is a power of 3, a point partition alone suffices: take $B_i = \varphi(A_i)$ and $\sigma_t = \varphi|_P$ (or any bijection with $\sigma_t(A_i) = B_i$).

3. Diameter 9 from $H(q)$

Corollary 3.1 (Theorem). Let $A_0, \dots, A_{k-1}$ be a partition of the points of $H(q)$ in which every A_i meets every plane π_p ($p \in P$), and $B_0, \dots, B_{k-1}$ a partition of the lines in which every B_i contains, for every line l, the line l or a line meeting l; suppose $|A_i| = |B_i|$. Then for every $r \geq 1$ there is a bipartite $(q+1+r)$-regular graph of diameter at most 9 with

$$2N(q)\,(2rk+1) \quad \text{vertices.}$$

The graph is Λ of Definition 2.1 with $I = I(q)$ (so $c = 5$), $\rho = 2$ and any bijections σ_t with $\sigma_t(A_i) = B_i$; Theorem 2.3 gives the diameter. Explicitly, the neighbours of (z, g), $z \in A_i$, are the $q+1$ vertices (l, g) with $l \ni z$ and the r vertices $(\sigma_t(z), g + (t-1)k + i + 1)$; the neighbours of (l, g), $l \in B_i$, are the $q+1$ vertices (p, g) with $p \in l$ and the r vertices $(\sigma_t^{-1}(l), g - (t-1)k - i - 1)$.

Proposition 3.2 (Theorem). In Corollary 3.1, $k \leq q^2 + q + 1$, with equality if and only if every A_i is an ovoid and every B_i is a spread. Hence $k = q^2 + q + 1$ needs $H(q)$ to have ovoids (they exist when q is a power of 3; $H(2)$ has none, Computer-checked 3.3; see §9 for what is known in general).

Proof. A point z lies in π_p exactly for the q^2+q+1 points $p \in \pi_z$. Since A_i meets all N planes, $|A_i|(q^2+q+1) \geq N$, i.e. $|A_i| \geq q^3 + 1$, and $\sum_i |A_i| = N$ gives $k \leq q^2 + q + 1$. Equality forces $|A_i| = q^3 + 1$ and every plane to meet A_i exactly once; then two points of A_i are never collinear (both would lie in π_z for one of them) and never at distance 4 (both would lie in π_y for the middle point y of the path), so A_i is an ovoid. The same for lines. Conversely, ovoids and spreads meet every 2-ball exactly once (§1). □

Computer-checked 3.3 (research/theorems/polylev/perfcodes.c, triples.py). $H(2)$ has no ovoid and exactly 28 spreads. $H(3)$ has exactly 6,318 spreads and 6,318 ovoids (exact-cover enumeration of the perfect codes of the 2-ball hypergraph); there are 3,411,720 disjoint pairs of spreads but **no three pairwise disjoint spreads**, so in particular no partition into 13 spreads, and $k = 13$ is impossible at $q = 3$. (The search for a partition was also run directly as an exact cover and found none.)

So the extremal case of Corollary 3.1 does not occur at $q = 3$, and near-extremal partitions are rare: dominating sets only slightly larger than ovoids pack badly. Partitions with k well below $q^2 + q + 1$ do exist:

Computer-checked 3.4 (local search, research/theorems/polylev/domlocal.c and translocal.c; every colouring was re-checked by the independent scorer domcheck.py, which tests every 2-ball).

q	side	k found	$q^2 + q + 1$	method
2	points	5	7	domlocal.c
3	both, balanced (line colouring = image of the point colouring under a duality of $H(3)$)	10	13	domlocal.c, dualfind.c
4	both, balanced	13	21	translocal.c, group T of order 13 (§3.5)
5	points	15	31	domlocal.c (squad r2/e11; $k = 16$ failed for two seeds, cost 6 and 7)
7	points	20	57	domlocal.c (squad r2/e12; $k = 24$ unresolved)
9	points	none at $k = 64$ (cost 555,007 after 600 s)	91	domlocal.c

"Points" means only the point side was searched; the line side is the same problem for the dual hexagon. The $q = 3$ colouring with $k = 10$ was lifted with $r = 1$ (lift9.py): 15,288 vertices, apsp: bipartite, 5-regular, diameter 9 (checks/d9_q3_k10_r1_D9.apsp.txt; the run with $D = 8$ fails). The fraction of $q^2 + q + 1$ reached falls as q grows, from 77% at $q = 3$ to 35% at $q = 7$, while the records need 79% at $q = 7$ and at least 49% anywhere.

Sizes (research/theorems/polylev/sizes.py). With k at its maximum $q^2 + q + 1$, and the smallest k that beats the lab's current value at the same $(\Delta, 9)$ (rows where even the maximum does not are omitted):

Delta	q	r	max k	order at max k	lab value	k needed to beat it
7	5	1	31	492,156	462,840	30 (97%)
9	7	1	57	4,509,840	3,499,776	45 (79%)
10	7	2	57	8,980,464	8,003,856	51 (89%)
10	8	1	73	11,010,006	8,003,856	53 (73%)
11	8	2	73	21,945,114	16,792,080	56 (77%)
11	9	1	91	24,313,380	16,792,080	63 (69%)
12	9	2	91	48,493,900	34,026,720	64 (70%)
13	9	3	91	72,674,420	67,742,688	85 (93%)
13	11	1	133	94,601,304	67,742,688	96 (72%)
14	11	2	133	188,848,296	116,068,656	82 (62%)
15	11	3	133	283,095,288	208,029,120	98 (74%)
15	13	1	183	295,239,756	208,029,120	129 (70%)
16	11	4	133	377,342,280	337,394,400	119 (89%)
16	13	2	183	589,675,044	337,394,400	105 (57%)
17	13	3	183	884,110,332	501,707,400	104 (57%)
18	13	4	183	1,178,545,620	790,626,816	123 (67%)
18	16	1	273	1,223,618,214	790,626,816	177 (65%)
19	13	5	183	1,472,980,908	1,218,394,320	152 (83%)
19	16	2	273	2,444,999,466	1,218,394,320	136 (50%)
19	17	1	307	1,855,575,540	1,218,394,320	202 (66%)
20	16	3	273	3,666,380,718	1,823,175,864	136 (50%)
20	17	2	307	3,708,133,884	1,823,175,864	151 (49%)

3.5 Group-equivariant colourings. Let $T = \langle h \rangle$ be a cyclic group of collineations of H of order d acting semiregularly on points and on lines, and fix a representative b_O of every T-orbit O. Given a block $a_O \in \{0, \dots, A-1\}$ and a shift $s_O \in \mathbb{Z}_d$ for every orbit, colour $h^j b_O$ by $a_O d + (j + s_O \bmod d)$. Then h maps the class of colour (a, c) onto the class $(a, c+1)$ and maps 2-balls to 2-balls, so a 2-ball $B_2(h^j x)$ sees every colour if and only if $B_2(x)$ does: it suffices to check one 2-ball per orbit, which shrinks the search by a factor d. If the numbers of orbits in the A blocks are the same on the point side and on the line side (the `fixed` mode of translocal.c), every colour class has as many points as lines, which is the balance condition of Corollary 3.1; and σ_t can be taken T-equivariant. We found (g2.py: linear collineations obtained by mapping the coordinate apartment onto random apartments, products of two of them, powers):

- **Computer-checked 3.6.** Cyclic groups of order $q^2 - q + 1$ acting semiregularly on the points and lines of $H(q)$ for $q = 3, 4, 7, 9$ (orders 7, 13, 43, 73; grp/T_q*.h, the 7×7 matrices; every line is mapped to a line). For $q = 5$ the group of order 21 has line orbits of size 7 (an element of order 3 fixes lines; $3 \mid q + 1$), so only its subgroup of order 7 is semiregular; the same happens for every $q \equiv 2 \pmod 3$.
- The Singer cycles of $SL(3, q) \le G_2(q)$, acting by $(v, a, w) \mapsto (gv, a, g^{-T} w)$ in the coordinates $x = (v, a, w)$, $v = (x_0, x_1, x_2)$, $a = x_3$, $w = (x_4, x_5, x_6)$, have order $q^2 + q + 1$; they act semiregularly when $q \not\equiv 1 \pmod 3$ (a fixed point would need an eigenvector of g over $\mathbb{F}_q$), and their subgroups of order prime to 3 act semiregularly for all q (grp/S*_q*.pperm: orders 7 and 13 at $q = 9$, 7 and 19 at $q = 11$, 61 at $q = 13$, 91 at $q = 16$).

4. Diameter 10: a one-sided ball lift and spread stars

For even diameter only one side needs colouring classes.

Definition 4.1. Let I, P, L, N, c be as in §2, let $\rho' \geq 1$ be odd, and let $L = C_0 \cup \cdots \cup C_{K-1}$ be a partition of the lines such that every C_i meets $\{l \in L : d(p,l) \leq \rho'\}$ for every point p. Let $r' \geq 1$, $m = 2r'K + 1$, and let $\sigma_{t,+}, \sigma_{t,-} : P \to L$ $(1 \leq t \leq r')$ be **arbitrary** bijections. Write $\iota(l) = i$ for $l \in C_i$. The graph Λ' has vertex set $(P \cup L) \times \mathbb{Z}_m$; the neighbours of (z,g), $z \in P$, are

$$(l,g)\ (l \ni z), \quad (\sigma_{t,+}(z),\ g+(t-1)K+\iota(\sigma_{t,+}(z))+1), \quad (\sigma_{t,-}(z),\ g-(t-1)K-\iota(\sigma_{t,-}(z))-1) \qquad (1 \leq t \leq r'),$$

and the neighbours of (l,g) are (p,g) $(p \in l)$, $(\sigma_{t,+}^{-1}(l), g-(t-1)K-\iota(l)-1)$ and $(\sigma_{t,-}^{-1}(l), g+(t-1)K+\iota(l)+1)$.

Theorem 4.2. Λ' is a simple bipartite graph on $2Nm$ vertices, each vertex has degree $\deg_I + 2r'$, and $\operatorname{diam} \Lambda' \leq \rho' + c + 2$.

Proof. The layer changes of the $2r'$ matching edges at a point are pairwise distinct and nonzero modulo m (they lie in $\pm[(t-1)K+1, tK]$), so Λ' is simple; it is bipartite with classes $P \times \mathbb{Z}_m$, $L \times \mathbb{Z}_m$. By the translations it suffices to bound $d((p,0),(l',h))$ for a point p and a line l'. If $h = 0$ this is $\leq c$. If $h = -((t-1)K+i+1)$, pick $l \in C_i$ with $d(p,l) \leq \rho'$; then $(p,0) \rightsquigarrow (l,0) \to (\sigma_{t,+}^{-1}(l), h) \rightsquigarrow (l',h)$ has length $\leq \rho'+1+c$. If $h = (t-1)K+i+1$, do the same with $\sigma_{t,-}$. So every point–line distance is $\leq \rho'+c+1$ (odd), and Lemma 2.2(b) gives the claim. □

Lemma 4.3 (Theorem). Let S be a spread of $H(q)$ and, for $s \in S$, let $C_s = B_2(s)$ (the line s and the lines meeting it). The C_s partition L, and every C_s contains a line at distance ≤ 3 from any given point.

Proof. The partition is the perfect-code property of spreads (§1). Let p be a point and $s \in S$; $d(p,s) \in \{1,3,5\}$. If $d(p,s) = 1$, take s. If $d(p,s) = 3$, the path $p\,m\,x\,s$ gives the line $m \ni p$ meeting s, at distance 1. If $d(p,s) = 5$, the path $p\,m_1\,x_1\,m_2\,x_2\,s$ gives $m_2 \in C_s$ at distance 3 from p. □

Corollary 4.4 (Theorem). For every prime power q and $r' \geq 1$ there is a bipartite $(q+1+2r')$-regular graph of diameter at most 10 with $2N(q)\,(2r'(q^3+1)+1)$ vertices (Theorem 4.2 with $I = I(q)$, $c = 5$, $\rho' = 3$, and the classes C_s of a Hermitian spread, $K = q^3+1$).

Computer-checked 4.5 (squad card r1/c14, research/squads/polylev/r1/c14/build_star.py; apsp.c). The graphs of Corollary 4.4 with $r' = 1$ and random bijections were built for $q = 2$ (2,394 vertices, degree 5) and $q = 3$ (41,496 vertices, degree 6), two seeds each; all-pairs BFS: bipartite, regular, diameter 8 ($q = 2$) and 9 ($q = 3$), within the bound 10.

The orders against the lab's current values at $D = 10$ (research/theorems/polylev/sizes.py):

Delta	q	matchings 2r'	K = q^3+1	order	lab value	ratio
5	2	2	9	2,394	95,472	0.025
6	3	2	28	41,496	524,160	0.079
7	2	4	9	4,662	2,298,240	0.002
7	4	2	65	357,630	2,298,240	0.156
8	3	4	28	82,264	8,043,840	0.010
8	5	2	126	1,976,436	8,043,840	0.246
9	2	6	9	6,930	23,975,280	0.000
9	4	4	65	712,530	23,975,280	0.030
10	3	6	28	123,032	60,141,120	0.002
10	5	4	126	3,945,060	60,141,120	0.066
10	7	2	344	27,019,824	60,141,120	0.449
11	4	6	65	1,067,430	148,729,680	0.007
11	8	2	513	76,920,246	148,729,680	0.517
12	5	6	126	5,913,684	325,832,976	0.018
12	7	4	344	54,000,432	325,832,976	0.166
12	9	2	730	194,108,460	325,832,976	0.596
13	8	4	513	153,765,594	677,714,400	0.227
14	7	6	344	80,981,040	1,326,627,120	0.061

Delta	q	matchings 2r'	K = q^3+1	order	lab value	ratio
14	9	4	730	388,084,060	1,326,627,120	0.293
14	11	2	1,332	944,241,480	1,326,627,120	0.712
15	8	6	513	230,610,942	2,467,618,800	0.093
16	9	6	730	582,059,660	4,220,872,800	0.138
16	11	4	1,332	1,888,128,648	4,220,872,800	0.447
16	13	2	2,198	3,537,245,796	4,220,872,800	0.838
18	11	6	1,332	2,832,015,816	11,787,972,672	0.240
18	13	4	2,198	7,073,687,124	11,787,972,672	0.600
19	16	2	4,097	18,331,903,590	19,233,722,496	0.953
20	13	6	2,198	10,610,128,452	30,518,778,360	0.348
20	17	2	4,914	29,656,019,484	30,518,778,360	0.972

None of these beats the lab's Haar graphs; the closest are $(19, 10)$ ($q = 16$, 95%) and $(20, 10)$ ($q = 17$, 97%). Why there is no record here: every class must meet the N sets $\{l : d(p, l) \le 3\}$ and every line lies in $(q+1)(q^2+1)$ of them, so $|C_i| \ge N/((q+1)(q^2+1))$ and $K \le (q+1)(q^2+1)$. The spread stars give $K = q^3 + 1$, within a factor $1 + 1/q$ of that bound, so better classes cannot help much (**Heuristic 4.6**: we did not prove that the bound is not attained). The loss is structural: each factor K costs two matchings (one per sign), against one per factor k at $D = 9$.

5. Diameter 11: disjoint lines and non-collinear points

Lemma 5.1 (Theorem). In $H(q)$, (a) the point set of any line meets $B_4(p)$ for every point p; (b) the pencil of any point z (the $q + 1$ lines through z) meets $B_4(l)$ for every line l.

Proof. (a) $d(p, \ell) \in \{1, 3, 5\}$; the point of ℓ on the (unique, by (F2)) shortest path from p to ℓ is at distance $d(p, \ell) - 1 \le 4$ from p. (b) is the dual statement: the line through z on the shortest path from l to z is at distance $d(l, z) - 1 \le 4$ from l. □

Corollary 5.2 (Theorem). Suppose $H(q)$ has K pairwise disjoint lines $\ell_0, \dots, \ell_{K-1}$ and K pairwise non-collinear points $z_0, \dots, z_{K-1}$. Then for every $r \ge 1$ there is a bipartite $(q + 1 + r)$-regular graph of diameter at most 11 with $2N(q)(2rK + 1)$ vertices.

Proof. Let A_i be the point set of ℓ_i and B_i the pencil of z_i ($|A_i| = |B_i| = q + 1$); they are pairwise disjoint and 4-dominating by Lemma 5.1. Distribute the remaining $N - K(q+1)$ points and lines evenly so that $|A_i| = |B_i|$ stays true (Remark 2.5) and apply Theorem 2.3 with $\rho = 4$, $c = 5$. □

Lemma 5.4 (Theorem). (a) Two points of $H(q)$ are at distance ≤ 4 in I if and only if they are orthogonal with respect to b. (b) Every totally singular line m of $Q(6, q)$ (a line of the quadric, not necessarily a line of H) contains a point at distance ≤ 4 from any given point p; so the $q+1$ points of m form a 4-dominating point set.

Proof. (a) Points at distance 2 lie on a common line of H, which is totally singular; points x, y at distance 4 have a common neighbour point z at distance 2 from both, so both lie in the totally singular plane π_z. In both cases $b(x, y) = 0$. Conversely, $p^\perp$ is the tangent hyperplane of $Q(6, q)$ at p (for q even, p is not the nucleus $\langle e_3 \rangle$ of b, so $p^\perp$ is still a hyperplane), and $p^\perp \cap Q(6, q)$ is a cone with vertex p over a parabolic quadric $Q(4, q)$: it has $1+q(q^3+q^2+q+1) = 1+q+q^2+q^3+q^4$ points. The points at distance ≤ 4 from p number $1+q(q+1)+q^3(q+1)$, the same; so the two sets coincide. (b) m is a 2-dimensional subspace and $p^\perp$ a hyperplane, so they share a nonzero vector, i.e. a point x of m with $b(p, x) = 0$; by (a), $d(p, x) \le 4$. □

Corollary 5.5 (Theorem). Suppose $Q(6, q)$ has K pairwise disjoint totally singular lines (a partial line spread of the quadric) and $H(q)$ has K pairwise non-collinear points. Then the conclusion of Corollary 5.2 holds: for every $r \ge 1$ there is a bipartite $(q + 1 + r)$-regular graph of diameter at most 11 with $2N(q)(2rK + 1)$ vertices. If q is a power of 3, the second hypothesis can be dropped: with a duality φ, the line classes $\varphi(m_i)$ (each a set of $q + 1$ lines of H) are 4-dominating (Remark 2.6 and Lemma 5.4(b)), pairwise disjoint and balanced with the point classes.

Proof. As for Corollary 5.2, with Lemma 5.4(b) for the point classes. □

Computer-checked 5.6. $Q(6,3)$: a randomized greedy finds 78 pairwise disjoint totally singular lines (85.7% of the bound 91; qspread.py); with the duality of $H(3)$ (research/d10D7/dual/dualfind.c) both sides have 78 classes, and the independent scorer domcheck.py confirms that every class meets every 4-ball. The lift with $r = 1$ (lift9.py) has 114,296 vertices; apsp: bipartite, 5-regular, diameter exactly 11 (checks/d11_q3_K78_r1_D11.apsp.txt; the run with $D = 10$ fails). Squad card r1/c13 built the pencil version at $q = 2$ ($K = 19$, the largest partial distance-2 spread of $H(2)$, found by exhaustive search): diameter 11 for $r = 1$ and 10 for $r = 2$.

5.7 First values at $(\Delta, 11)$, $\Delta \geq 11$. The 2013 table stops at $D = 10$ and the lab's beyond-table band covers $D = 11$ only for $\Delta \leq 10$, so no value is known to us for $(\Delta, 11)$, $11 \leq \Delta \leq 20$. Corollary 5.5 gives one as soon as K disjoint quadric lines and K non-collinear points (or, for $q = 9$, the duality) are exhibited. The sets below were found by squad round 2 and by the director. Every set was certified by research/theorems/polylev/d11cert.py (every class is the point set of a 2-dimensional totally singular subspace, classes pairwise disjoint; no line of $H(q)$ contains two of the chosen points); the squad sets were also certified by the finding card's own checker, and the $q = 8$ sets by the stand-alone verify_q8.py (Computer-checked 5.8). These graphs are far too large for a breadth-first check (up to $3.9 \cdot 10^{11}$ vertices); their diameter rests on Corollary 5.5, which was checked all-pairs at $q = 2, 3, 4$ (Computer-checked 5.6, 5.9).

$(\Delta, 11)$	q	r	K	order $2N(2rK+1)$	% of the bipartite Moore bound	lab value
(9, 11)	7	1	1,743	136,746,192	5.57	145,258,344
(10, 11)	8	1	3,292	**493,203,330**	6.29	445,709,568
(11, 11)	9	1	5,021	1,334,312,980	6.00	-
(12, 11)	9	2	5,021	2,668,493,100	4.68	-
(13, 11)	11	1	8,379	5,937,914,808	4.40	-
(14, 11)	11	2	8,379	11,875,475,304	3.98	-
(15, 11)	13	1	14,335	23,064,902,028	3.70	-
(16, 11)	13	2	14,335	46,128,999,588	3.73	-
(17, 11)	13	3	14,335	69,193,097,148	2.95	-
(18, 11)	16	1	29,087	130,135,264,350	3.04	-
(19, 11)	16	2	29,087	260,268,291,738	3.44	-
(20, 11)	16	3	29,087	390,401,319,126	3.02	-

(research/theorems/polylev/d11values.py; for each Δ the best (q, r) with $q + 1 + r = \Delta$ among the certified K.) The sets behind each K (research/theorems/polylev/certs/, re-checked together by certs/run_certs.sh, log certs/cert.log):

q	lines of $Q(6,q)$ (point classes)	points or duality (line classes)	K	% of $q^4 + q^2 + 1$
7	1,978 (squad r2/e09, ILS on all t.s. lines)	1,743 (orbit quotient, cyclic group of order 48)	1,743	71.1
8	3,292 (orbit quotient, order 63, then local search)	4,041 (orbit quotient, order 63)	3,292	79.1
9	5,021 (ILS on all t.s. lines, ils_lines.c of r2/e09, warm restart)	duality of $H(9)$	5,021	75.6

q	lines of $Q(6,q)$ (point classes)	points or duality (line classes)	K	% of q^4+q^2+1
11	9,975 (squad r2/e15, 75 orbits of a group of order 133)	8,379 (squad r2/e02)	8,379	56.8
13	15,536 (squad r2/e03)	14,335 (squad r2/e04)	14,335	49.9
16	32,391 (squad r2/e05, greedy rounds)	29,087 (orbit quotient, cyclic group of order 255; squad r2/e06 had 28,483 with order 91)	29,087	44.2

The percentages are lower bounds, not estimates of the maxima. (**Heuristic:** they fall with q at least partly because the searches had less time per element at larger q; we did not test this.)

Pairwise disjoint lines are a partial distance-2 spread and pairwise non-collinear points a partial distance-2 ovoid; both have at most $q^4+q^2+1 = N/(q+1)$ elements (each point lies on at most one of the lines; each line contains at most one of the points). Distance-2 ovoids attaining the bound are known only for small q (§9). The sizes, and the fraction of the bound that K must reach to beat the lab's current values at $(9,11)$ and $(10,11)$:

Delta	q	r	max K	order at max K	lab value	K needed to beat it
4	2	1	21	5,418	19,152	impossible
5	2	2	21	10,710	253,080	impossible
5	3	1	91	133,224	253,080	impossible
6	2	3	21	16,002	1,971,744	impossible
6	3	2	91	265,720	1,971,744	impossible
6	4	1	273	1,493,310	1,971,744	impossible
7	2	4	21	21,294	10,648,176	impossible
7	3	3	91	398,216	10,648,176	impossible
7	4	2	273	2,983,890	10,648,176	impossible
7	5	1	651	10,179,036	10,648,176	impossible
8	2	5	21	26,586	43,702,560	impossible
8	3	4	91	530,712	43,702,560	impossible
8	4	3	273	4,474,470	43,702,560	impossible
8	5	2	651	20,350,260	43,702,560	impossible
9	2	6	21	31,878	145,258,344	impossible
9	3	5	91	663,208	145,258,344	impossible
9	4	4	273	5,965,050	145,258,344	impossible
9	5	3	651	30,521,484	145,258,344	impossible
9	7	1	2,451	192,276,048	145,258,344	1852 (76%)
10	2	7	21	37,170	445,709,568	impossible
10	3	6	91	795,704	445,709,568	impossible
10	4	5	273	7,455,630	445,709,568	impossible
10	5	4	651	40,692,708	445,709,568	impossible
10	7	2	2,451	384,512,880	445,709,568	impossible
10	8	1	4,161	623,376,054	445,709,568	2975 (71%)
11	9	1	6,643	1,765,310,820	-	-
12	9	2	6,643	3,530,488,780	-	-

Delta	q	r	max K	order at max K	lab value	K needed to beat it
13	11	1	14,763	10,461,770,424	-	-
14	11	2	14,763	20,923,186,536	-	-
15	13	1	28,731	46,227,144,684	-	-
16	13	2	28,731	92,453,484,900	-	-
17	13	3	28,731	138,679,825,116	-	-
18	16	1	65,793	294,355,118,694	-	-
19	16	2	65,793	588,708,000,426	-	-
20	17	2	83,811	1,011,499,873,020	-	-

(Rows with "-": no lab value; the row shown is the largest order at that degree, at the upper bound for K.)

Computer-checked 5.3 (round 1, superseded by 5.8; squad cards r1/c11, r1/c12). Iterated local search on the whole collinearity graph gave 1,619 disjoint lines and 1,618 non-collinear points in $H(7)$, and 2,701 of each in $H(8)$ (unions of 37 orbits of the Singer subgroup of order 73 of $SL(3,8)$). Round 2 and the director raised these to the values in the tables above.

Computer-checked 5.8 (the value at $(10,11)$). Work in $\mathbb{F}_8 = \mathbb{F}_2[\omega]/(\omega^3+\omega+1)$ and write $a_0+a_1\omega+a_2\omega^2$ as the integer $a_0+2a_1+4a_2$. Let

$$h = \begin{pmatrix} 0 & 6 & 5 & 0 & 0 & 5 & 2 \\ 7 & 6 & 6 & 0 & 0 & 2 & 6 \\ 7 & 2 & 0 & 0 & 1 & 2 & 7 \\ 1 & 2 & 4 & 6 & 7 & 4 & 1 \\ 5 & 7 & 1 & 0 & 7 & 2 & 2 \\ 1 & 3 & 5 & 0 & 2 & 2 & 4 \\ 5 & 5 & 1 & 0 & 3 & 4 & 0 \end{pmatrix}$$

acting on column vectors. Then:

1. h maps every line of $H(8)$ to a line of $H(8)$ (it is a product of collineations found by g2.py; the induced line permutation was computed and checked line by line, lineorbquot.py), and $\langle h\rangle$ has order 63 on the points. Its point orbits: 594 of size 63, two of size 9, one of size 7, two fixed points.
2. **Points.** The union Z of 67 point orbits of $\langle h\rangle$ (64 of size 63, the orbit of size 7 and the two fixed points), 4,041 points, contains no two points on a common line of $H(8)$. The 67 orbit representatives are in research/theorems/polylev/certs/q8_point_reps.txt (one point per row, coordinates $X_0,\dots,X_6$).
3. **Lines.** The union $\mathcal{M}$ of 55 orbits of totally singular lines of $Q(6,8)$ (52 of size 63, one of size 7, two fixed lines), 3,285 lines, is a set of pairwise disjoint totally singular lines. Representatives (two spanning points per row) in certs/q8_line_reps.txt. Only 63 of the 3,285 lines are lines of $H(8)$; the rest are lines of the quadric that are not lines of the hexagon.
4. Hence (Corollary 5.5 with $K=3{,}285$, $r=1$, $m=6{,}571$) there is a bipartite 10-regular graph of diameter at most 11 on $2\cdot 37{,}449\cdot 6{,}571 = 492{,}154{,}758$ vertices. It beats the lab's 445,709,568 at $(10,11)$ by 10.4%.
5. **Local search adds 7 lines.** An iterated local search on all totally singular lines of $Q(6,8)$, started from $\mathcal{M}$ (research/theorems/polylev/d11q8/ils_lines_w.c, a warm-start version of squad card r2/e09's ils_lines.c), gives a set $\mathcal{M}'$ of 3,292 pairwise disjoint totally singular lines; it is no longer a union of orbits, and is listed in full in certs/q8_lines_3292_span.txt.gz (two spanning points per row). With $K=3{,}292$, $m=6{,}585$: $2\cdot 37{,}449\cdot 6{,}585 = 493{,}203{,}330$ **vertices**, 10.7% above the lab's value. With $r=2,3,4$ the same data give 986,331,762; 1,479,460,194; 1,972,588,626 at $(11,11)$–$(13,11)$, which the $q=9$ and $q=11$ rows of the table beat.

Checks: (a) certs/verify_q8.py, which uses no other file of the repository: it implements $\mathbb{F}_8$, applies h to the representatives, tests $Q=0$, tests every pair of the 4,041 points for orthogonality and, when orthogonal, for Tits' equations of §1 (no pair passes both), checks that every line is totally singular, and that the lines are pairwise disjoint (3,285 lines covering 29,565 distinct points; run again on the plain list $\mathcal{M}'$ with the identity matrix: 3,292 lines, 29,628 distinct points); (b) d11cert.py on the expanded sets against the hexagon file (hexgen2.c's $H(8)$; the two descriptions of $H(8)$ agree: 300 random lines of the hexagon file hex_q8.bin satisfy Tits' equations,

and the 3,222 lines of $\mathcal{M}$ that are not in that file do not). Both report $K = 3{,}285$ for $\mathcal{M}$ and $K = 3{,}292$ for $\mathcal{M}'$. (c) An independent checker written from scratch by a separate agent from §1 of this note and the certificate files only (certs/indep_check/check.py, output run.log): h permutes the 37,449 points of $Q(6,8)$ with order 63, $|Z| = 4{,}041$ with no collinear pair (its collinearity test gives every point exactly $q(q+1) = 72$ collinear points on a sample), $\mathcal{M}'$ is 3,292 pairwise disjoint totally singular lines (71 of them lines of $H(8)$), $K = 3{,}292$; seven negative controls (a perturbed h, powers of h of the wrong order, an added collinear point, an added non-singular point, two overlapping lines, a non-singular span) were all rejected.

The graph is explicit: certs/d11graph.py fixes the padding of the classes and σ_1 (both are free in Theorem 2.3; the file's docstring states the choice) and prints the neighbourhood of any vertex. Explicitly, with $A_i \supseteq m_i$ ($m_i \in \mathcal{M}$) and $B_i \supseteq$ the pencil of $z_i \in Z$: (p,g), $p \in A_i$, is adjacent to the 9 vertices (l,g) with $p \in l$ and to $(\sigma_1(p), g{+}i{+}1)$; (l,g), $l \in B_i$, to the 9 vertices (p,g) with $p \in l$ and to $(\sigma_1^{-1}(l), g{-}i{-}1)$, indices mod m ($m = 6{,}571$ for $\mathcal{M}$, 6,585 for $\mathcal{M}'$). The graphs have $4.9 \cdot 10^8$ vertices, too many for an all-pairs BFS; as a sample, for random pairs (half of them two copies of the same element in different layers) d11graph.py found $B_5(u) \cap B_6(v) \neq \emptyset$, i.e. distance ≤ 11, every time: 300 of 300 pairs for $\mathcal{M}$ and 200 of 200 for $\mathcal{M}'$; and $B_5(u) \cap B_5(v) = \emptyset$, i.e. distance exactly 11, for 42 and 32 of them (certs/q8_sample.log, certs/q8_sample_3292.log). It also checked that the sampled vertices have 10 distinct neighbours and that adjacency is symmetric. So the diameter is exactly 11; the upper bound rests on Corollary 5.5, the sample only illustrates it.

Computer-checked 5.9 (a small test of the same mix). At $q = 4$: 20 pairwise disjoint lines of $Q(6,4)$ that are not lines of $H(4)$ (traces: for opposite points a, b, the $q+1$ points collinear with b at distance 4 from a) and the pencils of 20 pairwise non-collinear points (both sets checked by d11cert.py). lift9.py with $r = 1$: 111,930 vertices; apsp: bipartite, 6-regular, diameter 10 (checks/d11_q4_ideal_r1_D11.apsp.txt), within the bound 11.

Computer-checked 5.10 (why pencils on the line side). For opposite lines L, M the trace L^M (the $q+1$ lines meeting L at distance 4 from M) is 4-dominating on lines for $q = 3$ (5 of 5 random traces) but not for $q = 4$ (240 lines are opposite to all $q+1$ trace lines) or $q = 5$ (800 lines), 5 random traces each, research/theorems/polylev/linetrace.py; point traces were 4-dominating in all cases, as Lemma 5.4(b) says. For $q = 3^h$ a duality maps point traces to line traces, so the line traces are 4-dominating by Remark 2.6; for other q we only have these small cases. Greedy completion of a trace to a 4-dominating line set needs $q-1$ further lines at $q = 4, 5$ (linetrace2.py), so these classes are much larger than pencils, and we used pencils.

Heuristic 5.11. The group of order $q^2 - 1$ is what made $q = 8$ work: its point quotient (572 admissible orbits) gave 97% of the bound, while the Singer group of order 73 gave 65% (37 orbits after 60 s of local search), order 57 gave 71% (52 orbits), and local search on the whole collinearity graph 70% (2,907 points, squad r2/e08). The same group type at $q = 7$ (order 48), $q = 9$ (order 80) and $q = 16$ (order 255) gave only 71%, 62% and 44% (local search on the quotients). We have no explanation, and we do not know whether $H(8)$ has a distance-2 ovoid (4,161 points).

Computer-checked 5.12 (exact searches on the orbit quotients, $q = 8$). With the SAT solver CaDiCaL (pysat; research/theorems/polylev/d11q8/ovoidsat.py), as an exact cover of the line orbits of $\langle h \rangle$ by point orbits: no $\langle h \rangle$-invariant set of pairwise non-collinear points contains 66 orbits of size 63 (UNSAT; the solver gives no proof certificate), and there is one with 65 (4,095 points, certs/q8_points_4095.txt.gz; d11cert.py: no line of $H(8)$ contains two of them). So an $\langle h \rangle$-invariant partial distance-2 ovoid has at most $65 \cdot 63 + 9 = 4{,}104$ points, and $H(8)$ has a partial distance-2 ovoid of 4,095 points (98.4% of 4,161). For lines (d11q8/spreadsat.py, same solver), no admissible line orbit of $\langle h \rangle$ covers the two fixed points or the orbit of size 7, so an $\langle h \rangle$-invariant partial line spread of $Q(6,8)$ has at most $(37{,}449 - 9)/9 = 4{,}160$ lines. Three instances were left undecided after 40 minutes each: an $\langle h \rangle$-invariant cover of all 594 point orbits of size 63 (66 line orbits), a packing with at least 58 line orbits of size 63, and a line spread invariant under the Singer group of order 73 (57 line orbits covering the 513 point orbits). Whether $Q(6,8)$ has a line spread we do not know (nothing in the sources of §9). $Q(6,2)$ has one (21 lines, one of them a line of $H(2)$; d11q8/smallspread.py, SAT on all totally singular lines); for $q = 3, 4$ the same run was undecided after 15 minutes.

For $\Delta \geq 11$ the lab has no values at $D = 11$ (the 2013 table stops at $D = 10$, and the lab's beyond-table band covers $D = 11$ only for $\Delta \leq 10$).

6. Diameter 8, quadrangles and octagons (remarks)

- $D = 8$**.** Theorem 4.2 with $\rho' = 1$ asks for classes of lines meeting every pencil, i.e. sets of lines covering every point; there are at most $q+1$ disjoint ones, so $m \le 2r'(q+1)+1$ layers at degree $q+1+2r'$. This is the regime of the lab's diameter-8 compound note (research/theorems/D8_COMPOUND.md, branch `claude/own-d8-compound-1z4e6s`: about $2q$ layers), and below the 2013 table's pattern at $D = 8$. No new cell.
- **Quadrangles.** For I the incidence graph of a generalized quadrangle of order (q, q) ($c = 3$), Theorem 2.3 gives diameter $5 + \rho$. At $\rho = 2$ ($D = 7$) a class must meet every $p^\perp$ (q^2+q+1 points), so $k \le q^2+q+1$ and the order is at most $2(q+1)(q^2+1)(2r(q^2+q+1)+1)$. For every $5 \le \Delta \le 20$ this bound is below the plain hexagon level graph $(2r+1) \cdot 2N(q)$ of the same degree (checked for all prime powers $q \le 19$; at $\Delta = 4$ it is 450 against the lab's 864). Mixing hexagon and quadrangle layers does not help in this framework: the bound $\rho + c + 2$ is set by the largest point–line distance c.
- **The Ree–Tits octagon** $GO(q, q^2)$, $q = 2^{2e+1}$, has incidence graph of diameter 8 with point degree q^2+1 and line degree $q+1$ and $|L|/|P| = (q^2+1)/(q+1)$. Balancing the degrees by duplicating points gives maximum degree $\ge q^2+1$: $\Delta = 5$ for $q = 2$ (the incidence graph itself, 4,680 vertices, diameter 8, against the table's 9,234 at $(5, 8)$) and $\Delta \ge 65$ for $q = 8$. Neither falls in a cell the lab tracks ($\Delta \le 20$), so octagons give nothing here.

7. Where this leaves diameters 8–10

- $D = 8$**.** No construction here reaches the lab's values or the 2013 table. The table's $D = 8$ entries are Gómez's compounds ΩH_q, which the lab found not to have diameter 8 as published (papers/Ozden_OmegaH_D8_note.pdf); the lab's own compound note gets about $2q$ layers, and §6 gives at most $2r'(q+1)+1$.
- $D = 9$**.** Corollary 3.1 is the first construction in this family whose order grows like $4rq^2 \cdot q^5$: with k at its maximum q^2+q+1 it would beat the lab's value at $D = 9$ for every $7 \le \Delta \le 20$ except $\Delta = 8$ (table in §3). The obstacle is the size of k that can actually be realised (Computer-checked 3.3, 3.4): perfect partitions do not exist at $q = 3$, Hermitian spreads of $H(9)$ pairwise intersect (Computer-checked 7.1), and the local searches reached only about half of the maximum. The record thresholds need $k/(q^2+q+1)$ between 49% ($(20, 9)$, $q = 17$) and 97%.
- $D = 10$**.** Corollary 4.4 holds for every q with no search at all; it reaches 84–97% of the lab's Haar values at $(16, 10)$, $(19, 10)$, $(20, 10)$ and beats none.
- $\Delta \ge 17$**.** The beyond-table cells at $D = 8$–10 hold Haar graphs at 4.5–7% of the Moore bound; the ball lifts reach that range only at $D = 9$ with large k (e.g. $(19, 9)$ and $(20, 9)$ need $k \ge 136$ at $q = 16$, half the maximum 273).

Computer-checked 7.1 (squad card r1/c02, research/squads/polylev/r1/c02/). In $H(9)$, 5,000 Hermitian spreads (lines of $H(9)$ in a hyperplane $v^\perp$ with q^3+1 lines of H) were sampled; every one of the 12,497,500 pairs meets, in 1 line (22.0%) or 10 lines (78.0%). One fixed Hermitian spread was compared with the line sets of all 597,871 hyperplanes of $PG(6, 9)$: the intersections have 1, 10 or 730 lines, never 0. Since $G_2(9)$ acts transitively on the elliptic hyperplanes (**Heuristic**: we did not check this transitivity by computer), no two Hermitian spreads of $H(9)$ are disjoint, and by Remark 2.6 the same holds for the Hermitian ovoids.

8. Computer checks

All files are under research/theorems/polylev/ unless stated otherwise; `hexgen2` is research/d10D7/hexgen2.c (the lab's generator of $H(q)$, which checks N points, N lines, $q+1$ points per line and lines per point), and `apsp` is research/lifts/coordinator_check/apsp.c (all-pairs bit-parallel BFS: simple, maximum degree, bipartite, connected, exact diameter; it assumes no symmetry).

claim	check	result
Theorem 2.3 / Cor. 3.1, $q = 3$, $k = 7$	lift9.py + apsp, $r = 1, 2, 3$ (10,920; 21,112; 31,304 vertices)	bipartite, regular of degree $q + 1 + r$; diameter 9, 9, 8; "$D = 8$" fails for $r = 1, 2$
Cor. 3.1, $q = 4$, $k = 13$	squad r1/c13: translocal colourings, lift9.py, apsp (73,710; 144,690 vertices)	diameter ≤ 9 for $r = 1, 2$; "$D = 8$" fails
Cor. 4.4, $q = 2, 3$, $r' = 1$	squad r1/c14: build_star.py + apsp (2,394; 41,496 vertices, 2 seeds each)	diameter 8 ($q = 2$), 9 ($q = 3$)
Cor. 5.2, 5.5	$q = 3$: 78 singular-line classes + duality, lift9.py + apsp (114,296 vertices); squad r1/c13: $q = 2$ pencils, $K = 19$	diameter 11 (and 10 fails); $q = 2$: diameter 11 ($r = 1$), 10 ($r = 2$)
Comp. 3.3	perfcodes.c (exact cover), triples.py	28 spreads, no ovoid in $H(2)$; 6,318 spreads and 6,318 ovoids in $H(3)$; no 3 pairwise disjoint spreads
Comp. 3.4	domlocal.c, translocal.c; independent scorer domcheck.py (tests every 2-ball by BFS)	see the table in §3
Comp. 3.6	g2.py (collineations by apartment maps, verified on all lines), singer_perms.py (orbit sizes printed)	semiregular groups listed in §3.5
Comp. 5.3	squad r1/c11, r1/c12 (ils_mis.c, mis2.c; sets re-verified)	$K \geq 1{,}618$ ($q = 7$), $K \geq 2{,}701$ ($q = 8$)
§5.7 table	certs/run_certs.sh: d11cert.py on every set (squad sets re-checked here; each card also ran its own checker)	$K = 1{,}743$; 3,292; 5,021; 8,379; 14,335; 29,087 for $q = 7$, 8, 9, 11, 13, 16
Comp. 5.8	certs/verify_q8.py (stand-alone, own $\mathbb{F}_8$ and Tits' equations), d11cert.py, certs/indep_check/check.py (separate author, with negative controls); d11graph.py (500 sampled pairs)	$K = 3{,}285$ (orbits), 3,292 (plus local search); sampled distances ≤ 11
Comp. 5.9	traces + pencils at $q = 4$, $K = 20$, lift9.py + apsp (111,930 vertices)	bipartite, 6-regular, diameter 10
Comp. 5.10	linetrace.py, linetrace2.py	line traces 4-dominating at $q = 3$ only (of 3, 4, 5)
Comp. 3.4, $q = 3$, $k = 10$	lift9.py + apsp, $r = 1$ (15,288 vertices)	diameter 9; "$D = 8$" fails
Comp. 7.1	squad r1/c02 (sample.py, full_check.py)	Hermitian spreads of $H(9)$ pairwise meet

The squad runs, their scoreboards and their logs are in research/squads/polylev/r1/ and r2/ (log of the rounds in research/squads/LOG.md).

9. Prior art

Searches by squad card r1/c16 (research/squads/polylev/r1/c16/prior_art.md, with evidence tags) and the lab's novelty file (research/lit/novelty-2026-10-25.md on branch `claude/novelty-literature-check-4i5frp`).

- **Level graphs ($\rho = 0$).** The $D = n + 1$ level graphs of generalized polygons with a tournament of layers are the Gómez–Fiol compounds $B(T)$ (J. Gómez, M. A. Fiol, *Dense compound graphs*, Ars Combin. 20-A (1985)) and give the 2013 table's diameter-7 column, credited to Delorme, Gómez and Quisquater (*On large bipartite graphs*, cited as submitted; not found). Remark 2.4 is that construction.
- **Voltage lifts of polygon incidence graphs at larger diameter.** C. Delorme, *Grands graphes de degré et diamètre donnés*, European J. Combin. 6 (1985) 291–302, §4: products $G * \Gamma'$ of the incidence graph G of a generalized quadrangle or hexagon with an oriented perfect matching; item (c), $G * C_{35}$, has degree $\Delta + 2$, diameter $D + 3$ and $35n$ vertices, i.e. for $H(q)$ degree $q + 3$, diameter 9 and $70N(q)$ vertices (card c16 read the full text; bipartiteness of (c) is not stated there; the bipartite versions are in C. Delorme, *Large bipartite graphs with given degree and diameter*, J. Graph Theory 9 (1985) 325–334, not read). This is the closest prior construction to Corollary 3.1, which has $2N(4k + 1)$ vertices at the same degree and diameter, more than $70N$ as soon as $k \geq 9$. Delorme's layers are indexed by a fixed cyclic group and every jump has a fixed voltage per arc; the ball lift lets the voltage depend on the colour class of the jumping element, which is what makes the number of layers grow like q^2.
- **Hubs and freed sets** (D7 note; R. Rajiv, Zenodo 2026, doi 10.5281/zenodo.23052995) concern diameter n and $n + 1$; not used here.
- **Ovoids, spreads, partitions.** Thas (J. Combin. Theory Ser. A 29 (1980)) relates ovoids of $H(q)$ to perfect codes (not read). Distance-2 ovoids of $H(q)$ are known only for $q = 2, 3, 4$ (De Wispelaere–Van Maldeghem 2008, cited by Bishnoi–Ihringer, arXiv:1606.07288). For partitions of the points of $H(q)$ into ovoids, of its lines into spreads, or the domatic number of its collinearity graph: nothing found in these sources.
- **Partial distance-2 ovoids and spreads.** Squad card r2/e13 found 273 pairwise non-collinear points in $H(4)$ by local search, a distance-2 ovoid; this is consistent with De Wispelaere–Van Maldeghem's distance-2 ovoid of $H(4)$ (we did not compare the two). For large partial distance-2 ovoids of $H(8)$ (Computer-checked 5.8: 4,041 of at most 4,161 points) and large partial line spreads of $Q(6, q)$, nothing was found in these sources.
- **Bipartite values at $D = 9$, $D = 11$ after 2013:** nothing found in these sources (the CombinatoricsWiki page could not be retrieved on 28 October 2026).